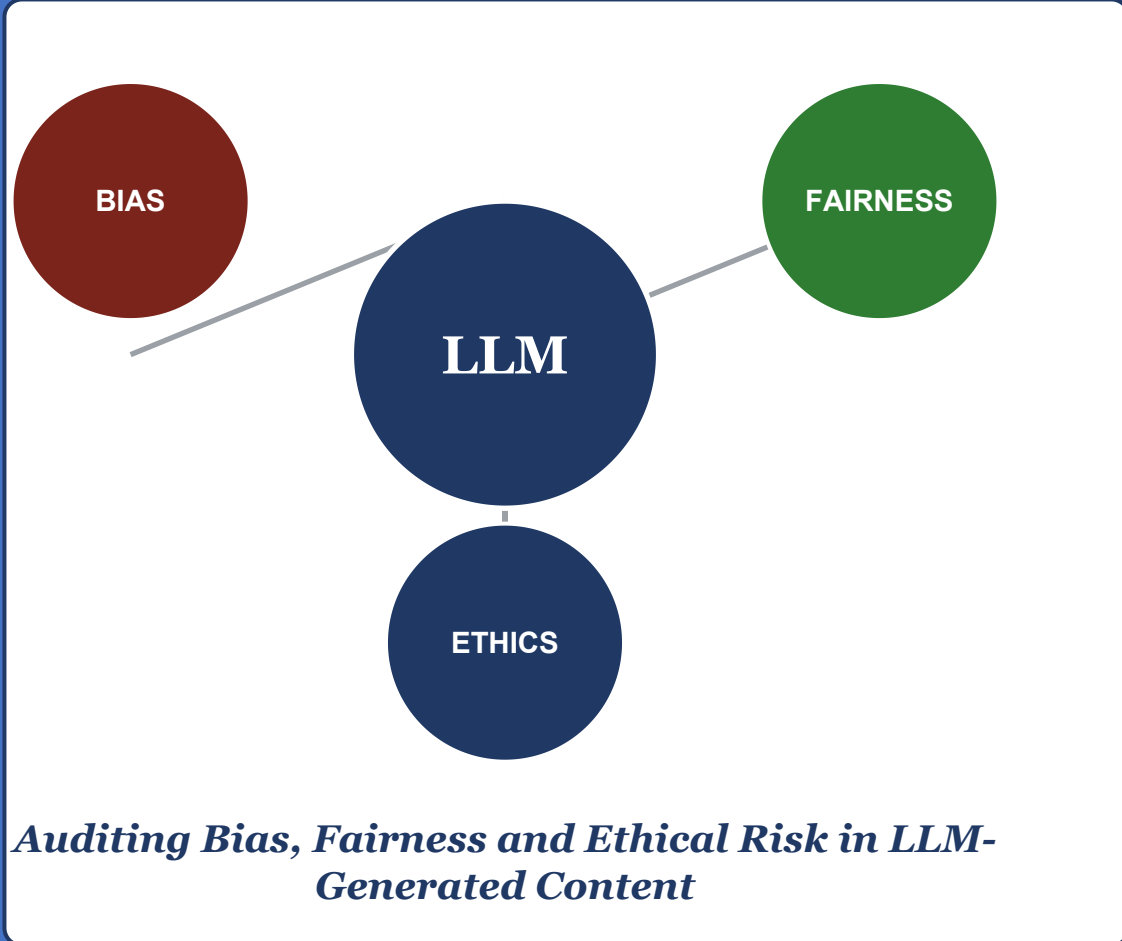


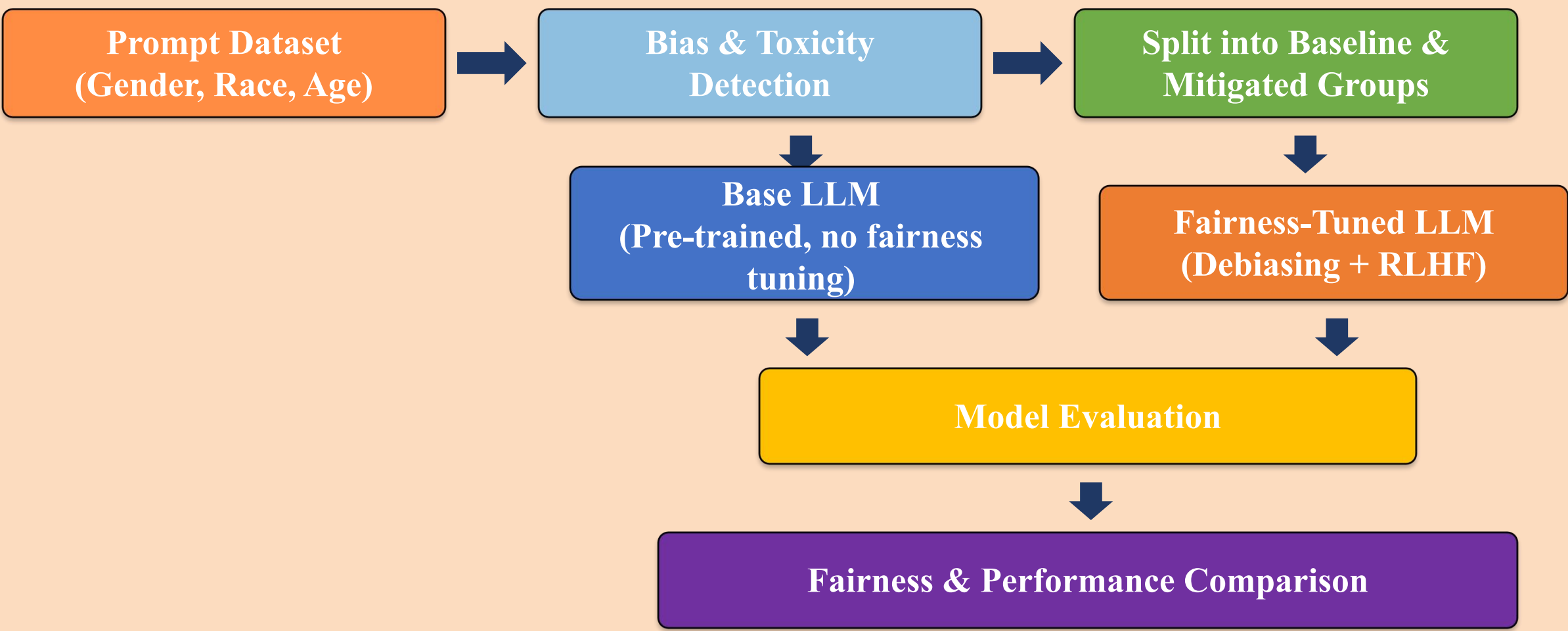
Responsible AI: Bias, Fairness and Ethics in Large Language Models

INTRODUCTION

- Large Language Models (LLMs) are increasingly deployed in high-stakes domains such as hiring, healthcare, lending, and criminal justice.
- Training corpora mirror historical and societal biases, which LLMs can absorb and amplify in their generated outputs.
- Fairness concerns include stereotyping, representational harm, and unequal treatment across gender, race, and age groups.
- This study benchmarks bias in base LLMs against fairness-tuned variants using standardized prompt sets and fairness metrics.
- Goal: identify mitigation strategies that meaningfully reduce bias while preserving model accuracy and fluency.



MATERIALS AND METHODS



RESULTS

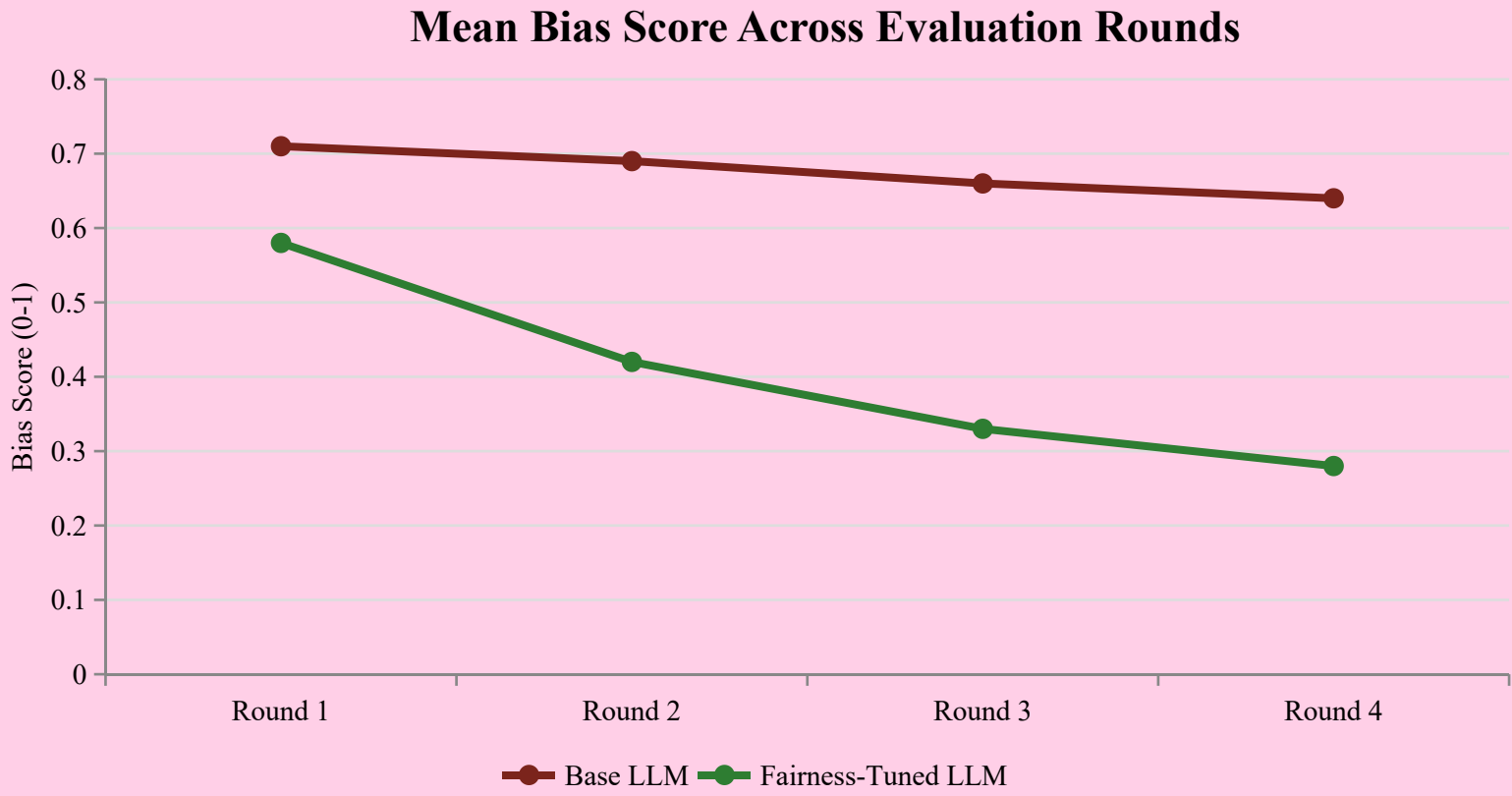


Fig 1: Bias score decreases faster for the fairness-tuned model across rounds.

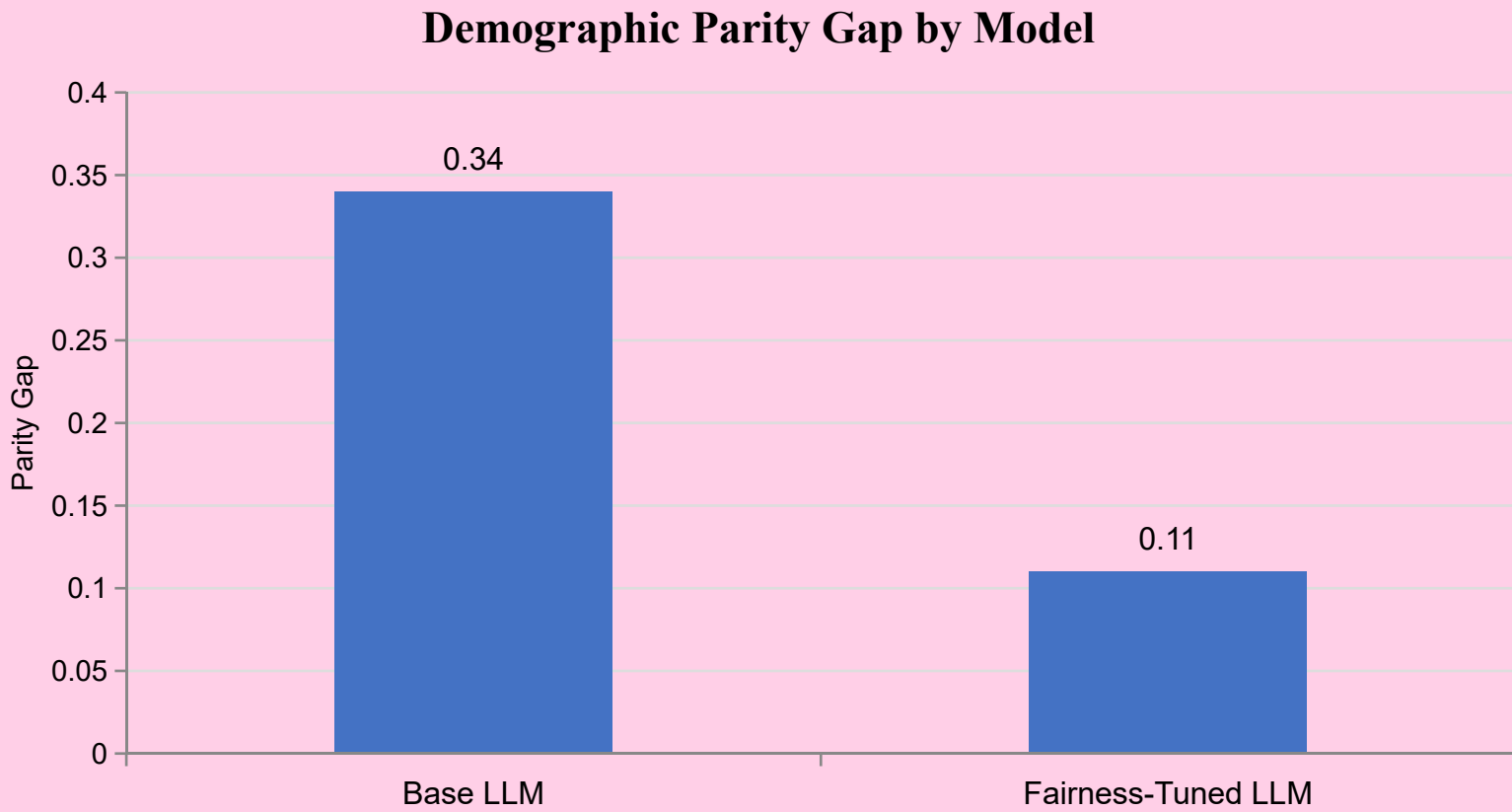


Fig 2: Fairness tuning reduces the demographic parity gap by roughly 68%.

DISCUSSION AND CONCLUSION

- Fairness-tuned LLMs cut stereotypical bias substantially with only a small drop in task accuracy.
- Base LLMs showed statistically significant bias against non-Western names in hiring-style prompts ($p < 0.001$).
- A trade-off persists between strict fairness constraints and output fluency / diversity.
- Debiasing at the RLHF stage was more effective than post-hoc output filtering alone.
- Future scope: multilingual fairness audits and real-time bias-monitoring dashboards for deployed LLMs.

Group Statistics: Base vs Fairness-Tuned LLM

Group	N	Mean Bias Score	Std. Deviation	Std. Error Mean
Base LLM	40	0.68	0.142	0.0224
Fairness-Tuned LLM	40	0.31	0.098	0.0155

Independent-samples t-test: $t(78) = 13.42$, $p < 0.001$, indicating a statistically significant reduction in mean bias score after fairness tuning.

BIBLIOGRAPHY

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